

PitchCoach AI

Cost Implication Analysis

Infrastructure, Operations & Margin Analysis

CONFIDENTIAL

Prepared for Automagikal

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Executive Summary

This document provides a comprehensive cost analysis for the PitchCoach AI platform, covering all infrastructure components, AI model usage, payment processing fees, and operational expenses. The analysis is designed to inform pricing strategy and ensure sustainable margins at the target price point of \$15-20 per month per user.

Category	Monthly Fixed Cost	Cost Per Active User	Notes
Infrastructure	\$44-74	\$0.02-0.05	Scales with storage
AI Models	\$0	\$0.15-2.50	Per-session usage
Payment Processing	\$0	\$0.45-0.60	3.4-3.9% per transaction
TOTAL	\$44-74	\$0.62-3.15	Highly variable by usage tier

Table 1: Cost Summary Overview

1. Infrastructure Costs (Fixed Monthly)

Infrastructure costs represent the baseline operational expenses required to run the platform regardless of user count. These costs include database hosting, authentication services, file storage, video streaming, and application hosting. The following breakdown details each component with specific pricing tiers.

Service	Provider	Plan	Monthly Cost	Limits
Database	Supabase	Pro	\$25	8GB, 250GB bandwidth
Authentication	Clerk	Pro	\$25	10K MAUs included
Hosting	Vercel	Pro	\$20	100GB bandwidth
File Storage	Cloudflare R2	Pay-as-you-go	\$0-15	\$0.015/GB storage
Video Streaming	Cloudflare Stream	Pay-as-you-go	\$0-10	\$5/1000 min stored
TOTAL FIXED			\$44-74	Scales with storage usage

Table 2: Monthly Fixed Infrastructure Costs

2. AI Model Costs (Variable Per-Session)

AI model costs are the primary variable expense, directly tied to user engagement with each coaching module. These costs scale linearly with usage and represent the most significant margin consideration for pricing strategy.

2.1 Model Pricing Reference

Model	Provider	Input (per 1M tokens)	Output (per 1M tokens)	Best For
Claude Sonnet 4	Anthropic	\$3.00	\$15.00	Text analysis, E1/E2/E4
Gemini 2.5 Flash	Google	\$0.10	\$0.40	Video analysis, E3
Gemini 1.5 Pro	Google	\$1.25	\$5.00	Long video analysis, E4

Table 3: AI Model Pricing (as of March 2025)

3. Module-by-Module Cost Breakdown

3.1 E1: Pitch Deck Analyser

The Deck Analyser processes uploaded PDF or PPTX files through Claude Sonnet for comprehensive content and visual analysis. Each deck analysis requires document extraction, multi-page processing, and structured scoring output.

Cost Component	Unit	Cost	Notes
File Upload (R2)	Per upload	\$0.0001	Negligible
PDF Extraction	Per deck	\$0.002	Server-side processing
Claude Sonnet Input	~5K tokens/deck	\$0.015	Deck content + images
Claude Sonnet Output	~2K tokens/deck	\$0.030	Analysis results
TOTAL PER DECK		\$0.047	~5 cents per analysis

Table 4: E1 Deck Analyser Cost Per Session

3.2 E2: Elevator Pitch Script Coach

The Script Coach analyzes text input or uploaded documents, evaluating the 5-element structure (Hook, Problem, Solution, Credibility, CTA) and generating rewritten versions with alternative opening hooks.

Cost Component	Unit	Cost	Notes
Claude Sonnet Input	~1K tokens/script	\$0.003	Script + prompts

Claude Sonnet Output	~1.5K tokens/script	\$0.023	Analysis + rewrites
TOTAL PER SCRIPT		\$0.026	~2.6 cents per analysis

Table 5: E2 Script Coach Cost Per Session

3.3 E3: Live Pitch Coach (Video)

The Live Pitch module processes video recordings up to 3 minutes for delivery and body language analysis. Gemini 2.5 Flash is used for its cost-effective video understanding capabilities.

Cost Component	Unit	Cost	Notes
Video Upload (Stream)	Per video	\$0.05	Storage + processing
Gemini 2.5 Flash Input	~3 min video	\$0.02	Video + audio analysis
Gemini 2.5 Flash Output	~2K tokens	\$0.001	Analysis results
TOTAL PER VIDEO		\$0.071	~7 cents per analysis

Table 6: E3 Live Pitch Cost Per Session

3.4 E4: Full Pitch Session (30-min Video + Deck)

The Full Session module is the most resource-intensive, combining 30-minute video analysis with deck evaluation. Gemini 1.5 Pro is required for longer video context, combined with Claude Sonnet for deck analysis.

Cost Component	Unit	Cost	Notes
Deck Analysis (Claude)	Per deck	\$0.047	Same as E1
Video Upload (Stream)	Per video	\$0.50	30 min storage
Gemini 1.5 Pro Input	~30 min video	\$1.50	Long context processing
Gemini 1.5 Pro Output	~4K tokens	\$0.020	Comprehensive analysis
Synthesis (Claude)	Combined analysis	\$0.10	Merge deck + video results
TOTAL PER SESSION		\$2.17	Most expensive module

Table 7: E4 Full Session Cost Per Session

4. Module Cost Comparison

Module	Cost Per Session	Sessions at \$15/mo	Sessions at \$20/mo
E1: Deck Analyser	\$0.047	319 sessions	425 sessions
E2: Script Coach	\$0.026	576 sessions	769 sessions
E3: Live Pitch	\$0.071	211 sessions	281 sessions
E4: Full Session	\$2.17	6 sessions	9 sessions

Table 8: Breakeven Session Count by Module

5. Payment Processing Costs

Payment processing fees vary by provider and region. PitchCoach AI supports both Stripe (global markets) and Pystack (African markets) to optimize costs for different customer bases.

Provider	Markets	Fee Structure	\$15 Subscription Cost	\$20 Subscription Cost
Stripe	Global (USD/EUR/GBP)	2.9% + \$0.30	\$0.74 (4.9%)	\$0.88 (4.4%)
Pystack	Africa (NGN/ZAR/KES/GHS)	1.5% + \$0.10	\$0.33 (2.2%)	\$0.40 (2.0%)

Table 9: Payment Processing Fees by Provider

6. User Scenario Analysis

This section models cost implications across different user engagement patterns to understand margin distribution and identify optimal pricing tiers.

User Type	Monthly Sessions	Variable Cost	Margin @ \$15	Margin @ \$20
Light User (E1/E2 only)	5 sessions	\$0.37	\$13.89 (93%)	\$18.63 (93%)
Moderate User (Mixed)	10 sessions (mix)	\$0.80	\$13.46 (90%)	\$18.20 (91%)
Active User (All modules)	20 sessions (mix)	\$2.50	\$11.76 (78%)	\$16.50 (83%)
Power User (E4 heavy)	5 E4 sessions	\$10.85	\$3.41 (23%)	\$8.15 (41%)

Table 10: User Scenario Cost Analysis

7. Pricing Recommendations

Based on the cost analysis, the following pricing structure is recommended to ensure sustainable margins while maintaining competitive positioning in the market.

Plan	Price	E1 Limit	E2 Limit	E3 Limit	E4 Limit	Est. Margin
Free Trial	\$0	1	1	0	0	N/A
Starter	\$9/mo	5	10	2	0	70-85%
Professional	\$19/mo	20	50	10	2	75-90%
Enterprise	\$49/mo	Unlimited	Unlimited	30	10	65-80%

Table 11: Recommended Pricing Tiers

8. Scale Projections (1000 Users)

Projected monthly costs at 1000 monthly active users with average usage patterns of 8 E1, 10 E2, 3 E3, and 1 E4 session per user per month.

Cost Category	Calculation	Monthly Cost
Fixed Infrastructure	Supabase + Clerk + Vercel + Storage	\$74
E1: Deck Analyses (8000)	8,000 x \$0.047	\$376
E2: Script Coach (10000)	10,000 x \$0.026	\$260
E3: Live Pitch (3000)	3,000 x \$0.071	\$213
E4: Full Session (1000)	1,000 x \$2.17	\$2,170
Payment Processing	1000 x \$15 x 4.5%	\$675
TOTAL MONTHLY COST		\$3,768
REVENUE (1000 users)	1000 x \$15	\$15,000
NET MARGIN		\$11,232 (75%)

Table 12: 1000-User Scale Projection

9. Key Findings & Recommendations

- E4 is the margin risk: At \$2.17 per session, E4 full sessions consume 58% of variable costs. Limit E4 access in lower tiers.
- E1/E2 are highly profitable: At ~3-5 cents per session, deck and script analysis offer 90%+ margins.
- Payment processing varies by region: Paystack saves 2.7% vs Stripe for African markets. Route users accordingly.
- Infrastructure scales well: Fixed costs of \$44-74/month become negligible above 500 users (\$0.09/user).
- Recommended launch price: \$19/month for Professional tier maximizes margin while staying competitive.
- Freemium strategy: Free trial with 1 E1 + 1 E2 session costs ~\$0.07 per acquisition - highly sustainable.